VIET LINH NGUYEN

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Professional Summary

Results-driven **AI Engineer** with a proven track record of architecting and deploying production-grade intelligent systems. Recognized for deep expertise in designing complex **Multi-agent architectures** (LangGraph/LangChain), implementing **Real-time AI Audio/Video pipelines** (WebRTC/LiveKit), and orchestrating advanced RAG ecosystems with vector databases (pgvector). Highly proactive in bridging the gap between cutting-edge LLM research and scalable engineering practices to deliver resilient, context-aware applications that solve complex real-world challenges while optimizing token usage and cloud infrastructure.

Technical Skills

Languages: Python, JavaScript/TypeScript (Node.js), Go (Golang), SQL.
AI & LLM Tech: Multi-agent Systems (LangGraph, LangChain), Real-time Multimodal AI, RAG Pipelines, Hybrid Search, LiveKit, Whisper, Prompt Engineering.
Frameworks & Tools: WebRTC, FastAPI, Next.js, React, Docker, Git, Linux, CI/CD.
Databases & Infra: PostgreSQL, pgvector, Vector Databases, Redis, MongoDB, RESTful/gRPC.

Experience

Swork Joint Stock Company (Swork JSC) — Hanoi, Vietnam
Full-Stack / AI Engineer — *Dec 2025 – Mar 2026*

- **Real-Time Meeting Intelligence Architecture:** Spearheaded the development of a low-latency meeting assistant by integrating **LiveKit** (WebRTC streaming) and **OpenAI Whisper**, enabling real-time audio transcription with sub-second latency across distributed online meetings.
- **Advanced Context-Aware NLP Pipelines:** Architected and deployed an automated LLM workflow utilizing **Vector Databases** for semantic retrieval. Engineered highly optimized prompts to accurately synthesize meeting minutes, extract critical action items, and generate contextual summaries.
- **High-Concurrency System Operations & Scaling:** Designed the scalable end-to-end backend architecture using **Go (Golang)** and frontend with **React**. Optimized Voice Activity Detection (VAD) algorithms and implemented asynchronous processing queues, significantly improving system stability and throughput under heavy concurrent traffic.

Education

Hanoi University of Industry (HaUI) — Hanoi, Vietnam
Bachelor of Information Technology, GPA: 3.36 / 4.0 — *2022 – 2026*

- **Achievements**: 2nd Prize in Student Scientific Research (2024–2025) with the project "Bitter".

Personal Projects

ATrips – AI-Powered Smart Travel Planner | *LangChain, Next.js, PostgreSQL, PostGIS* Dec 2025 – Mar 2026

- **5-Layer Multi-Agent Architecture:** Architected and engineered complex orchestration flows from initial intent classification to comprehensive result synthesis using **LangChain**. Developed highly flexible fallback mechanisms and caching strategies, successfully optimizing LLM token costs while maintaining 99% conversational accuracy.
- **46+ Autonomous AI Tools Integration:** Deployed an extensive, automated tool-calling ecosystem capable of autonomously booking flight tickets, reserving accommodations, and retrieving real-time weather forecasts. Designed a rigorous 5-stage execution pipeline (filtering, rate-limiting, edge-caching, secure execution, and payload distillation) to seamlessly interface with external third-party travel APIs.
- **Real-Time Context Management & SSE Streaming:** Implemented Server-Sent Events (**SSE**) streaming to deliver instantaneous, responsive AI feedback to users. Built an intelligent document ingestion engine capable of parsing complex formats (PDF, DOCX) with advanced token compression algorithms, ensuring the model maintains deep, long-context awareness across extended planning sessions.
- **Geospatial Recommendation Engine:** Engineered a highly personalized Points of Interest (POI) recommendation system leveraging user preferences and historical data. Managed complex geospatial querying across 82 interconnected Prisma models using **PostgreSQL** and **PostGIS**, dramatically accelerating location-based retrieval speeds.

Lumen – AI Literature Review Assistant | *FastAPI, LangGraph, pgvector, Next.js* 2026

- **Advanced LangGraph Architecture:** Designed a highly complex state machine encompassing 7 specialized agent nodes (*query_planner, search_agent, language_bias_checker, matrix_extraction, gap_analysis, conflict_detection, review_writer*) to autonomously navigate and control the entire research automation lifecycle.
- **Map-Reduce & Multi-Query Retrieval:** Implemented asynchronous Map-Reduce patterns to synthesize research gaps across hundreds of scientific documents. Built a sophisticated dual-context engine combining knowledge graph traversal with semantic chunk-based retrieval to maximize LLM response accuracy and depth.
- **Citation-Safe Report Synthesis:** Developed robust, high-throughput pipelines for PDF full-text extraction and LLM-based text normalization. Engineered a dedicated citation validation agent that meticulously cross-references generated reports against source chunks, ensuring 100% hallucination-free output.
- **Multi-Provider Hybrid Search Engine:** Integrated flexible AI provider adapters (Anthropic, OpenAI, DeepSeek) seamlessly tied to **pgvector** for highly scalable, cross-document vector storage and hybrid semantic searching operations.

InterviewX – AI Mock Interviewer (Top 30 Google Hackathon) | *LiveKit, Gemini, FastAPI, Next.js* 2026

- **Ultra-Low Latency Streaming Architecture:** Engineered a real-time, voice-to-voice AI interview platform leveraging **LiveKit** (WebRTC) and Google's **Gemini 2.5 Flash Native Audio** model. Achieved seamless, human-like conversational dynamics with instantaneous responses.
- **Distributed Real-time Processing Pipeline:** Orchestrated bidirectional audio streaming architectures utilizing a decoupled Python worker service. Implemented robust Voice Activity Detection (VAD) and proactive interruption management to simulate natural human conversational latency and turn-taking.
- **Automated CV & JD Analysis Engine:** Developed asynchronous background job runners to automatically parse and evaluate candidate CVs against complex Job Descriptions (JDs), dynamically generating highly contextualized interview scripts and opening questions.
- **Dynamic Contextual Memory & Live Dashboarding:** Designed real-time state management systems to retain conversational history across long sessions. Built a live HR dashboard in **Next.js** to continuously stream and monitor interview transcripts, behavioral metrics, and AI evaluations on the fly.